

69. (a) Laguerre's differential equation

$$ty'' + (1 - t)y' + ny = 0$$

is known to possess polynomial solutions when n is a nonnegative integer. These solutions are naturally called **Laguerre polynomials** and are denoted by $L_n(t)$. Find $y = L_n(t)$, for $n = 0, 1, 2, 3, 4$ if it is known that $L_n(0) = 1$.

(b) Show that

$$\mathcal{L}\left\{\frac{e^t}{n!} \frac{d^n}{dt^n} t^n e^{-t}\right\} = Y(s),$$

where $Y(s) = \mathcal{L}\{y\}$ and $y = L_n(t)$ is a polynomial solution of the DE in part (a). Conclude that

$$L_n(t) = \frac{e^t}{n!} \frac{d^n}{dt^n} t^n e^{-t}, \quad n = 0, 1, 2, \dots$$

Computer Lab Assignments

70. In this problem you are led through the commands in *Mathematica* that enable you to obtain the symbolic Laplace transform of a differential equation and the solution of the initial-value problem by finding the inverse transform. In *Mathematica* the Laplace transform of a function $y(t)$ is obtained using **LaplaceTransform** [$y[t], t, s$]. In line two of the syntax, we

replace **LaplaceTransform** [$y[t], t, s$] by the symbol **Y**. (If you do not have *Mathematica*, then adapt the given procedure by finding the corresponding syntax for the CAS you have on hand.)

Consider the initial-value problem

$$y'' + 6y' + 9y = t \sin t, \quad y(0) = 2, \quad y'(0) = -1.$$

Precisely reproduce and then, in turn, execute each line in the given sequence of commands. Either copy the output by hand or print out the results.

diffequat = $y''[t] + 6y'[t] + 9y[t] == t \sin[t]$

transformdeq = **LaplaceTransform** [**diffequat**, **t**, **s**]/.

{y[0] -> 2, y'[0] -> -1,

LaplaceTransform [**y[t], t, s**] -> **Y}**

soln = **Solve**[**transformdeq**, **Y**] // **Flatten**

Y = **Y**/. **soln**

InverseLaplaceTransform[**Y**, **s**, **t**]

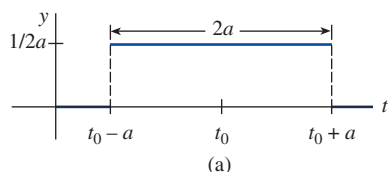
71. Appropriately modify the procedure of Problem 70 to find a solution of

$$y''' + 3y' - 4y = 0, \quad y(0) = 0, \quad y'(0) = 0, \quad y''(0) = 1.$$

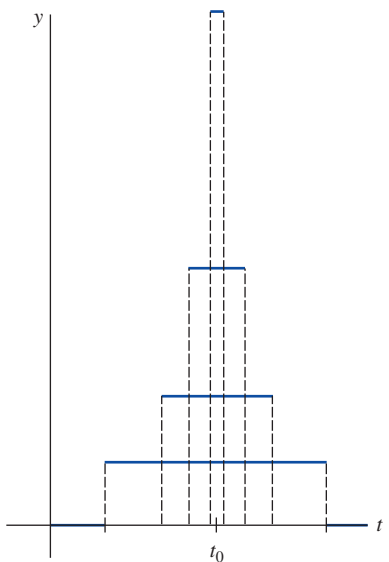
72. The charge $q(t)$ on a capacitor in an *LC*-series circuit is given by

$$\frac{d^2 q}{dt^2} + q = 1 - 4\mathcal{U}(t - \pi) + 6\mathcal{U}(t - 3\pi), \quad q(0) = 0, \quad q'(0) = 0.$$

Appropriately modify the procedure of Problem 70 to find $q(t)$. Graph your solution.



(a)



(b) Behavior of δ_a as $a \rightarrow 0$

FIGURE 4.5.1 Unit impulse

4.5 The Dirac Delta Function

INTRODUCTION Just before the *Remarks* on page 223, we indicated that as an immediate consequence of Theorem 4.2.3, $F(s) = 1$ cannot be the Laplace transform of a function f that is piecewise continuous on $[0, \infty)$ and of exponential order. In the discussion that follows we are going to introduce a function that is very different from the kinds that you have studied in previous courses. We shall see that there does indeed exist a function, or more precisely a *generalized* function, whose Laplace transform is $F(s) = 1$.

Unit Impulse Mechanical systems are often acted on by an external force (or emf in an electrical circuit) of large magnitude that acts only for a very short period of time. For example, a vibrating airplane wing could be struck by lightning, a mass on a spring could be given a sharp blow by a ball peen hammer, a ball (baseball, golf ball, tennis ball) could be sent soaring when struck violently by some kind of club (baseball bat, golf club, tennis racket). The graph of the piecewise-defined function

$$\delta_a(t - t_0) = \begin{cases} 0, & 0 \leq t < t_0 - a \\ \frac{1}{2a}, & t_0 - a \leq t < t_0 + a \\ 0, & t \geq t_0 + a, \end{cases} \quad (1)$$

$a > 0$, $t_0 > 0$, shown in **FIGURE 4.5.1(a)**, could serve as a model for such a force. For a small value of a , $\delta_a(t - t_0)$ is essentially a constant function of large magnitude that is “on” for just a very short period of time, around t_0 . The behavior of $\delta_a(t - t_0)$ as $a \rightarrow 0$ is illustrated in Figure 4.5.1(b). The function $\delta_a(t - t_0)$ is called a **unit impulse** since it possesses the integration property $\int_0^\infty \delta_a(t - t_0) dt = 1$.

The Dirac Delta Function In practice it is convenient to work with another type of unit impulse, a “function” that approximates $\delta_a(t - t_0)$ and is defined by the limit

$$\delta(t - t_0) = \lim_{a \rightarrow 0} \delta_a(t - t_0). \quad (2)$$

The latter expression, which is not a function at all, can be characterized by the two properties

$$(i) \quad \delta(t - t_0) = \begin{cases} \infty, & t = t_0 \\ 0, & t \neq t_0 \end{cases} \quad \text{and} \quad (ii) \quad \int_0^{\infty} \delta(t - t_0) dt = 1.$$

The unit impulse $\delta(t - t_0)$ is called the **Dirac delta function**.

It is possible to obtain the Laplace transform of the Dirac delta function by the formal assumption that $\mathcal{L}\{\delta(t - t_0)\} = \lim_{a \rightarrow 0} \mathcal{L}\{\delta_a(t - t_0)\}$.

Theorem 4.5.1 Transform of the Dirac Delta Function

For $t_0 > 0$, $\mathcal{L}\{\delta(t - t_0)\} = e^{-st_0}. \quad (3)$

PROOF: To begin, we can write $\delta_a(t - t_0)$ in terms of the unit step function by virtue of (11) and (12) of Section 4.3:

$$\delta_a(t - t_0) = \frac{1}{2a} [\mathcal{U}(t - (t_0 - a)) - \mathcal{U}(t - (t_0 + a))].$$

By linearity and (14) of Section 4.3, the Laplace transform of this last expression is

$$\mathcal{L}\{\delta_a(t - t_0)\} = \frac{1}{2a} \left[\frac{e^{-s(t_0 - a)}}{s} - \frac{e^{-s(t_0 + a)}}{s} \right] = e^{-st_0} \left(\frac{e^{sa} - e^{-sa}}{2sa} \right). \quad (4)$$

Since (4) has the indeterminate form 0/0 as $a \rightarrow 0$, we apply L'Hôpital's rule:

$$\mathcal{L}\{\delta(t - t_0)\} = \lim_{a \rightarrow 0} \mathcal{L}\{\delta_a(t - t_0)\} = e^{-st_0} \lim_{a \rightarrow 0} \left(\frac{e^{sa} - e^{-sa}}{2sa} \right) = e^{-st_0}. \quad \equiv$$

Now when $t_0 = 0$, it seems plausible to conclude from (3) that

$$\mathcal{L}\{\delta(t)\} = 1.$$

The last result emphasizes the fact that $\delta(t)$ is not the usual type of function that we have been considering, since we expect from Theorem 4.2.3 that $\mathcal{L}\{f(t)\} \rightarrow 0$ as $s \rightarrow \infty$.

EXAMPLE 1 Two Initial-Value Problems

Solve $y'' + y = 4\delta(t - 2\pi)$ subject to

(a) $y(0) = 1, \quad y'(0) = 0$ (b) $y(0) = 0, \quad y'(0) = 0.$

The two initial-value problems could serve as models for describing the motion of a mass on a spring moving in a medium in which damping is negligible. At $t = 2\pi$ the mass is given a sharp blow. In part (a) the mass is released from rest 1 unit below the equilibrium position. In part (b) the mass is at rest in the equilibrium position.

SOLUTION (a) From (3) the Laplace transform of the differential equation is

$$s^2 Y(s) - s + Y(s) = 4e^{-2\pi s} \quad \text{or} \quad Y(s) = \frac{s}{s^2 + 1} + \frac{4e^{-2\pi s}}{s^2 + 1}.$$

Using the inverse form of the second translation theorem, (15) of Section 4.3, we find

$$y(t) = \cos t + 4 \sin(t - 2\pi) \mathcal{U}(t - 2\pi).$$

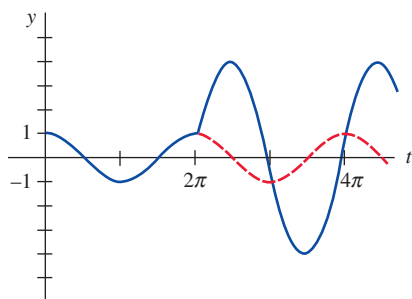


FIGURE 4.5.2 In Example 1(a), moving mass is struck at $t = 2\pi$

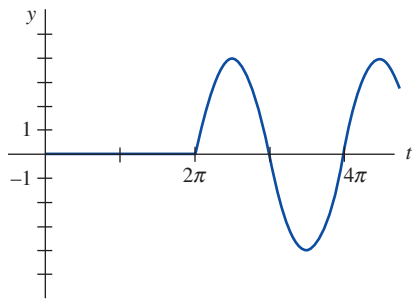


FIGURE 4.5.3 In Example 1(b), mass is at rest until struck at $t = 2\pi$

Since $\sin(t - 2\pi) = \sin t$, the foregoing solution can be written as

$$y(t) = \begin{cases} \cos t, & 0 \leq t < 2\pi \\ \cos t + 4 \sin t, & t \geq 2\pi. \end{cases} \quad (5)$$

In **FIGURE 4.5.2** we see from the graph of (5) that the mass is exhibiting simple harmonic motion until it is struck at $t = 2\pi$. The influence of the unit impulse is to increase the amplitude of vibration to $\sqrt{17}$ for $t > 2\pi$.

(b) In this case the transform of the equation is simply

$$Y(s) = \frac{4e^{-2\pi s}}{s^2 + 1},$$

and so

$$\begin{aligned} y(t) &= 4 \sin(t - 2\pi) \mathcal{U}(t - 2\pi) \\ &= \begin{cases} 0, & 0 \leq t < 2\pi \\ 4 \sin t, & t \geq 2\pi. \end{cases} \end{aligned} \quad (6)$$

The graph of (6) in **FIGURE 4.5.3** shows, as we would expect from the initial conditions, that the mass exhibits no motion until it is struck at $t = 2\pi$. ≡

REMARKS

(i) If $\delta(t - t_0)$ were a function in the usual sense, then property (ii) on page 249 would imply $\int_0^\infty \delta(t - t_0) dt = 0$ rather than $\int_0^\infty \delta(t - t_0) dt = 1$. Since the Dirac delta function did not “behave” like an ordinary function, even though its users produced correct results, it was met initially with great scorn by mathematicians. However, in the 1940s Dirac’s controversial function was put on a rigorous footing by the French mathematician Laurent Schwartz in his book *Théorie des distributions*, and this, in turn, led to an entirely new branch of mathematics known as the **theory of distributions** or **generalized functions**. In this theory, (2) is not an accepted definition of $\delta(t - t_0)$, nor does one speak of a function whose values are either ∞ or 0. Although we shall not pursue this topic any further, suffice it to say that the Dirac delta function is best characterized by its effect on other functions. If f is a continuous function, then

$$\int_0^\infty f(t) \delta(t - t_0) dt = f(t_0) \quad (7)$$

can be taken as the *definition* of $\delta(t - t_0)$. This result is known as the **sifting property** since $\delta(t - t_0)$ has the effect of sifting the value $f(t_0)$ out of the set of values of f on $[0, \infty)$. Note that property (ii) (with $f(t) = 1$) and (3) (with $f(t) = e^{-st}$) are consistent with (7).

(ii) In the *Remarks* in Section 4.2 we indicated that the transfer function of a general linear n th-order differential equation with constant coefficients is $W(s) = 1/P(s)$, where $P(s) = a_n s^n + a_{n-1} s^{n-1} + \cdots + a_0$. The transfer function is the Laplace transform of function $w(t)$, called the **weight function** of a linear system. But $w(t)$ can be characterized in terms of the discussion at hand. For simplicity let us consider a second-order linear system in which the input is a unit impulse at $t = 0$:

$$a_2 y'' + a_1 y' + a_0 y = \delta(t), \quad y(0) = 0, \quad y'(0) = 0.$$

Applying the Laplace transform and using $\mathcal{L}\{\delta(t)\} = 1$ shows that the transform of the response y in this case is the transfer function

$$Y(s) = \frac{1}{a_2 s^2 + a_1 s + a_0} = \frac{1}{P(s)} = W(s) \quad \text{and so} \quad y = \mathcal{L}^{-1}\left\{\frac{1}{P(s)}\right\} = w(t).$$

From this we can see, in general, that the weight function $y = w(t)$ of an n th-order linear system is the zero-state response of the system to a unit impulse. For this reason $w(t)$ is called as well the **impulse response** of the system.